

AeroCement Build Guide

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Triple-utility passive solar-thermal concrete system

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What Is AeroCement?

AeroCement transforms ordinary concrete into **multifunctional energy infrastructure**:

1. ✨ **Solar Capture** - Absorbs 95–98% of solar radiation
2. 📦 **Thermal Storage** - Stores heat for 7+ nights underground
3. 🏠 **Structural Material** - Load-bearing walls

Performance: ~12.91 kWh/m² nightly | 7-night accumulation = ~82.98 kWh

Quick Start

Clone the repo

```
git clone [https://github.com/jesseray718/aerocement]
(https://github.com/jesseray718/aerocement)
cd aerocement

# Read the build guide
cat docs/BUILD_GUIDE.md

# Calculate your system size
python3 aerocement/calc/thermal_calculator.py
```

Key Documents

Document	Description
BUILD_GUIDE.md	Step-by-step construction
MIX_DESIGN.md	Concrete recipes
THERMAL_PERFORMANCE.md	Performance data
BOM.md	Bill of materials

How It Works

1. Solar Capture

- Open-cell aerated concrete with zirconium
- Volumetric blackbody: 95–98% absorption
- No separate solar panels needed

2. Heat Transfer

- Heated air moves through underground channels
- Thermal labyrinth stores energy in ground battery
- Decouples thermal mass from living space

3. Energy Recovery

- Stirling engine or heat exchanger
- ~3.11 kW continuous from 10m² array
- Zero fuel cost, zero emissions

System Sizing

Building Size	Panel Area	Storage Volume	Est. Cost
Small (500 sq ft)	10 m ²	5 m ³	\$800–\$1,200
Medium (1,000 sq ft)	20 m ²	10 m ³	\$1,500–\$2,200
Large (2,000 sq ft)	40 m ²	20 m ³	\$2,800–\$4,000

Performance Data

Location: Sikeston, Missouri (37.13N, 89.53W)

Insolation: 5.2 kWh/m²/day average

Measured: 12.91 kWh/m² daily capture

Storage efficiency: 85–92%

Discharge: 7–10 nights

Round-trip: 78–84%

☐ Main Site

<https://jessera718.github.io> - Full ecosystem docs

☐ Related Projects

Project	Description
OpenRoot	Ferrocement domes + thermal labyrinths
Wisdom Scaffold	Ancient wisdom + computation
Renaissance Protocol	Proof-of-work for physical labor

☐ Community

- **Discussions:** <https://github.com/jessera718/aerocement/discussions>
 - **Issues:** <https://github.com/jessera718/aerocement/issues>
 - **Permies Forum:** <https://permies.com/t/379092>
 - **Email:** jrm8908@proton.me
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💰 Support This Work

AeroCement is **open-source hardware** (CC-BY-SA-4.0):

- ✓ Build it yourself - All designs free
 - 💰 Sponsor: <https://github.com/sponsors/jessera718>
 - ☐ Share your builds in Discussions
 - ☐ Submit improvements via PR
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☐ License

Hardware & Docs: CC-BY-SA-4.0

Software: GPL-3.0

You can: Share, Adapt, Use commercially
Requirements: Attribution + ShareAlike

Full: <https://creativecommons.org/licenses/by-sa/4.0/>

☐ Start Here

1. Read [BUILD_GUIDE.md](#)
 2. Join [Discussions](#)
 3. Watch for video tutorials (coming soon)
 4. [Sponsor](#) for early access
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Last updated: September 2026 | Version: 1.0.0

🏠 Community Builds

Have you built AeroCement? Share your project!

- Post photos in [Discussions](#)
- Tag [@jessera718](#) on social media
- Get featured in the README

Community builds coming soon - be the first!